



Course Objective and Outcome Form

Department of Electrical and Computer Engineering

School of Engineering and Physical Sciences

North South University, Bashundhara, Dhaka-1229, Bangladesh

1. **Course Number and Title:** CSE 311 Database Management Systems CSE 311L
Database Management Systems Laboratory
2. **Number of Credits:** 3+0
3. **Type:** Core
4. **Prerequisites:** CSE 225
5. **Contact Hours:** 3 hours (theory) + 3 hours (lab)
6. **Instructor:** Farhan Tanvir Khan
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Office Hours: ST 9:40 AM - 2:30 PM, RA 9:40 AM - 11:10 AM
7. **Course Summary:** This course introduces students with database management systems for the first time in their undergraduate study. Drawbacks of flat file system are demonstrated and advantages of relational database systems are introduced. The course examines the logical organization of databases: the entity-relationship model; the hierarchical, network, and relational data models and their languages. Functional dependencies and normal forms are discussed. Design, implementation, and optimization of query languages; security and integrity; concurrency control, different levels of indices, e.g., tree and hash based indices are introduced. Access costs are compared for different alternatives. This course has separate mandatory laboratory sessions every week in a separate course CSE 311L which has 0 credits, but the students (in group) use hands on SQL queries and as a culmination, they build a full fledged database system including a front end. The evaluation of the lab works is carried over to the theory part of the course.
8. **Course Objectives:**
The objectives of this course are:
 1. To make students comprehend the advantages of using a database system over flat files.
 2. To get students familiar with requirement analysis specially data requirements of an organization
 3. To introduce the conceptual design from requirement analysis using E-R diagrams and also mapping ER diagrams into relational schema.
 4. To introduce the basics and usage of relational algebra that are the foundation of SQL.
 5. To transform a relational design into physical database design using popular commercialized databases, e.g., Oracle, MySQL etc.

6. To demonstrate and show the evils of redundancy by introducing the concepts of functional dependencies and their types.
7. To design full-fledged physical database systems with least redundancy and most optimized manner.
8. To build their independent projects emphasizing the data requirement.

9. Course Outcomes (COs):

Upon Successful completion of this course, students will be able to:

Sl.	CO Description	Weightage (%)
1.	explain conceptual model through entity relationship diagrams (E-R Diagrams) and extended ER Diagrams	30
2.	use relational algebra in simple and complex queries based on set theory	20
3.	construct code in SQL DDL (Data Definition Language) and SQL DML (Data Manipulation Language) for table creation and query processing	25
4.	build a Web-based relational database system using scripting languages (e.g. PHP) and an open-source database development system (e.g. MySQL)	25

10. Mapping of CO-PO:

Sl.	CO Description	Program Outcome	Bloom's taxonomy domain/level (C: Cognitive P: Psychomotor or A: Affective)	Delivery methods and activities	Assessment tools
CO1	Explain conceptual model through entity relationship diagrams (E-R diagram) and extended ER	a	C2	Lectures, Notes	Quiz, Exam

	diagram. Build hierarchy and inheritance through specialization/generalization (Extended ER diagram (EER)). Translate conceptual model (ER and EER diagrams) to relational schema with different alternatives				
CO2	Use relational algebra in simple and complex queries based on set theory	a	C3	Lectures, Notes	Quiz, Exam
CO3	Construct code in SQL DDL (Data Definition Language) and SQL DML (Data Manipulation Language) for table creation and query processing	c	C3	Lectures, Notes, Lab	Quiz, Exam
CO4	Build a Web-based relational database system, using scripting languages (e.g. PHP) and an open-source database development system (e.g. MySQL)	c	C6	Lectures, Notes	Report and Demonstration

11. Resources

Text books:

No	Name of Author(s)	Year of Publication	Title of Book	Edition	Publisher's Name	ISBN
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1	Ramez Elmasri, Shamkant B. Navathe	2015	Fundamentals of Database Systems	7 th	Pearson	ISBN-13: 978-0133 97 0777
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Reference books:

N o	Name of Author(s)	Year of Publication	Title of Book	Edition	Publisher's Name	ISBN
1	Abraham Silberschatz, Henry F. Korth,	2011	Database System Concepts	6 th	McGraw-Hill	ISBN-13: 978-0073 52 3323

	S. Sudarshan					
2	Raghu Ramakrishnan, Johannes Gehrke	2003	Database Management System	3 rd	McGraw-Hill	ISBN-13: 978-0072 46 5631
3	Timothy Boronczyk, Elizabeth Narmore, Jason Gerner <i>et al.</i>	2009	Beginning PHP 6, Apache, MySQL 6 Web Development	1 st	Wiley	ISBN-13: 978-0470 39 1143

12. Weightage Distribution among Assessment Tools:

Theory

Assessment Tools	Weight (%)
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Quizzes	20
Midterm	30
Final Exam	30
Lab Work	10
Term Project	10
Total	100

13. Grading policy: As per NSU grading policy available in

<http://www.northsouth.edu/academic/grading-policy.html>